FIFA PLAYERS

	name	full_name	birth_date	age	height_cm	weight_kgs	positions	nationality	overall_rating	potential	...	long_shots	aggression	interceptions	pos
0	L. Messi	Lionel Andrés Messi Cuccittini	6/24/1987	31	170.18	72.1	CF,RW,ST	Argentina	94	94	...	94	48	22	
1	C. Eriksen	Christian Dannemann Eriksen	2/14/1992	27	154.94	76.2	CAM,RM,CM	Denmark	88	89	...	89	46	56	
2	P. Pogba	Paul Pogba	3/15/1993	25	190.50	83.9	CM,CAM	France	88	91	...	82	78	64	
3	L. Insigne	Lorenzo Insigne	6/4/1991	27	162.56	59.0	LW,ST	Italy	88	88	...	84	34	26	
4	K. Koulibaly	Kalidou Koulibaly	6/20/1991	27	187.96	88.9	CB	Senegal	88	91	...	15	87	88	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
17949	R. McKenzie	Rory McKenzie	10/7/1993	25	175.26	74.8	RM,CAM,CM	Scotland	67	70	...	54	69	41	
17950	M. Sipľak	Michal Sipľak	2/2/1996	23	182.88	79.8	LB	Slovakia	59	67	...	22	62	55	
17951	J. Bekkema	Jan Bekkema	4/9/1996	22	185.42	89.8	GK	Netherlands	59	67	...	9	27	10	
17952	A. Al Yami	Abdulrahman Al Yami	6/19/1997	21	175.26	64.9	ST,LM	Saudi Arabia	59	71	...	58	38	15	
17953	Júnior Brumado	José Francisco dos Santos Júnior	5/15/1999	19	190.50	79.8	ST	Brazil	59	75	...	53	67	20	

17954 rows × 51 columns

This table contains a dataset of **17,954 football players** with **51 attributes** for each player. It includes basic information such as player name, full name, birth date, age, height, weight, position, and nationality. The dataset also provides performance-related metrics like **overall rating**, **potential**, and various football skills.

Here is explanation for each column:

name Short name of the player.

full_name Player's full name.

birth_date Date of birth.

age Player's age.

height_cm Height in centimeters.

weight_kgs Weight in kilograms.

positions Playing positions (e.g., ST, CAM, CB).

nationality Country of nationality.

overall_rating Current overall skill rating.

potential Maximum potential rating the player can reach.

value_euro Estimated market value in euros (€).

wage_euro Weekly wage/salary in euros (€).

preferred_foot Dominant foot (Left or Right).

international_reputation(1-5) Global reputation level (1 = low, 5 = world-class).

weak_foot(1-5) Ability to use the weaker foot.

skill_moves(1-5) Skill move rating.

body_type Physical body build/type.

release_clause_euro Buyout/release clause amount in euros.

national_team National team represented by the player.

national_rating Rating within the national team context.

national_team_position Position played in the national team.

national_jersey_number Jersey number for the national team.

crossing Ability to deliver accurate crosses.

finishing Ability to score goals from chances.

heading_accuracy Accuracy when heading the ball.

short_passing Accuracy of short passes.

volleys Ability to strike airborne balls.

dribbling Ball control while moving past opponents.

curve Ability to curve the ball during shots or passes.

freekick_accuracy Accuracy of free kicks.

long_passing Accuracy of long-distance passes.

ball_control General control of the ball.

acceleration Speed of reaching top pace.

sprint_speed Maximum running speed.

agility Ability to change direction quickly.

reactions Speed of responding to situations.

balance Ability to stay stable while moving.

shot_power Strength behind shots.

jumping Jumping ability.

stamina Endurance throughout a match.

strength Physical strength in challenges.

long_shots Ability to score from long distances.

aggression Intensity in physical and defensive play.

interceptions Ability to intercept passes.

positioning Ability to find effective positions on the field.

vision Ability to spot passing opportunities.

penalties Penalty kick-taking ability.

composure Calmness under pressure.

marking Ability to track and mark opponents.

standing_tackle Effectiveness of standing tackles.

sliding_tackle Effectiveness of sliding tackles.

```
<class 'pandas.DataFrame'>
RangeIndex: 17954 entries, 0 to 17953
Data columns (total 51 columns):
 #   Column                         Non-Null Count  Dtype
---  ------                         --------------  -----
 0   name                           17954 non-null  str
 1   full_name                      17954 non-null  str
 2   birth_date                     17954 non-null  str
 3   age                            17954 non-null  int64
 4   height_cm                      17954 non-null  float64
 5   weight_kgs                     17954 non-null  float64
 6   positions                      17954 non-null  str
```

```
 41  long_shots                     17954 non-null  int64
 42  aggression                     17954 non-null  int64
 43  interceptions                  17954 non-null  int64
 44  positioning                    17954 non-null  int64
 45  vision                         17954 non-null  int64
 46  penalties                      17954 non-null  int64
 47  composure                      17954 non-null  int64
 48  marking                        17954 non-null  int64
 49  standing_tackle                17954 non-null  int64
 50  sliding_tackle                 17954 non-null  int64
dtypes: float64(7), int64(35), str(9)
memory usage: 7.0 MB
```

This output provides a quick overview of the dataset structure. It shows:

The dataset type (pandas.DataFrame).

Number of rows (17,954 entries).

Number of columns (51 columns).

Column names.

Number of non-null values in each column.

Data type of each column (int64, float64, str).

Total memory usage (7.0 MB).

	age	height_cm	weight_kgs	overall_rating	potential	value_euro	wage_euro	international_reputation(1-5)	weak_foot(1-5)	skill_moves(1-5)	...	long
count	17954.000000	17954.000000	17954.000000	17954.000000	17954.000000	1.769900e+04	17708.000000	17954.000000	17954.000000	17954.000000	...	17954.0
mean	25.565445	174.946921	75.301047	66.240169	71.430935	2.479280e+06	9902.134628	1.111674	2.945695	2.361034	...	46.8
std	4.705708	14.029449	7.083684	6.963730	6.131339	5.687014e+06	21995.593750	0.392168	0.663691	0.763223	...	19.4
min	17.000000	152.400000	49.900000	47.000000	48.000000	1.000000e+04	1000.000000	1.000000	1.000000	1.000000	...	3.0
25%	22.000000	154.940000	69.900000	62.000000	67.000000	3.250000e+05	1000.000000	1.000000	3.000000	2.000000	...	32.0
50%	25.000000	175.260000	74.800000	66.000000	71.000000	7.000000e+05	3000.000000	1.000000	3.000000	2.000000	...	51.0
75%	29.000000	185.420000	79.800000	71.000000	75.000000	2.100000e+06	9000.000000	1.000000	3.000000	3.000000	...	62.0
max	46.000000	205.740000	110.200000	94.000000	95.000000	1.105000e+08	565000.000000	5.000000	5.000000	5.000000	...	94.0

8 rows × 42 columns

This table includes measures such as count, mean, standard deviation, minimum, maximum, and quartiles (25%, 50%, and 75%).

This summary helps understand the distribution, and range of numerical features before conducting deeper analysis.

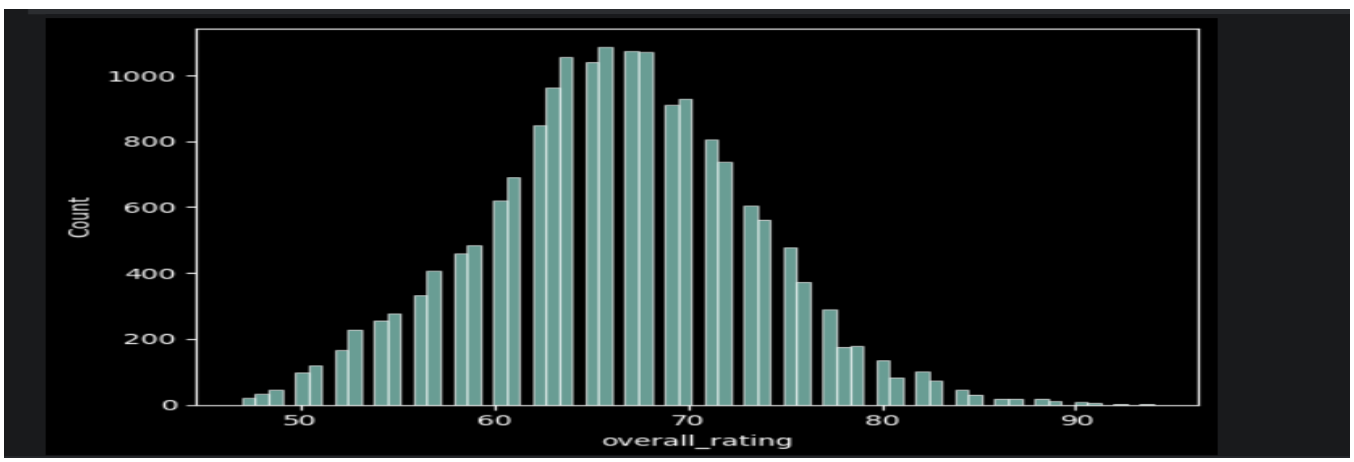
For several variables such as age, overall rating, and potential, the mean and 50% values are very close.

```
curve              0
long_passing       0
ball_control       0
acceleration       0
sprint_speed       0
agility            0
reactions          0
balance            0
stamina            0
strength           0
vision             0
dtype: int64
```

```
age                0
positions          0
overall_rating     0
potential          0
value_euro       255
crossing           0
finishing          0
short_passing      0
dribbling          0
curve              0
long_passing       0
ball_control       0
```

The dataset was cleaned by removing columns with low analytical importance, such as personal and less relevant attributes.
This helped reduce noise and improve the quality of the analysis.

After cleaning, most of the remaining features contained no missing values, which indicates that the dataset is well-structured and suitable for further analysis.
The `value_euro` column had missing values (255 rows). Since this feature is important for market value analysis, the missing values can be filled using the median.



This histogram shows the distribution of player overall_rating :

The majority of players have ratings between **60 and 75**, showing this is the core population of average-to-good players.

Very few players are below **50 or above 85**, indicating that extremely weak and elite players are rare.

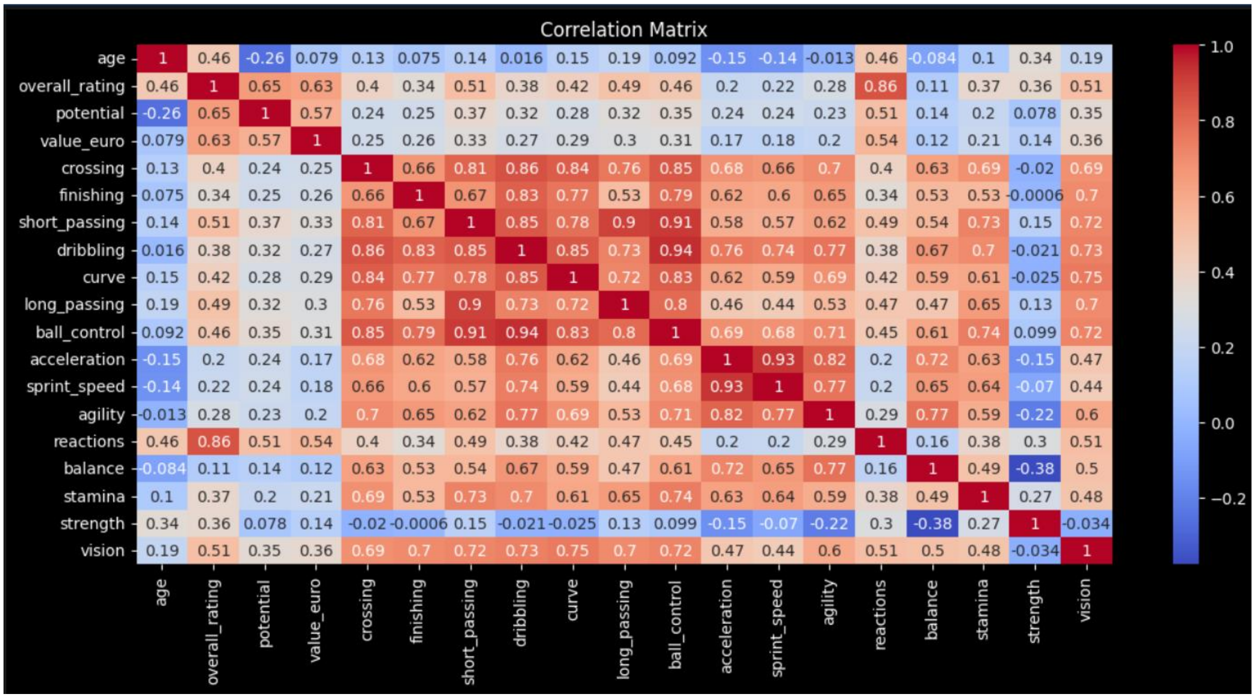
The boxplot shows several outliers.

These extreme values increase the spread of the data and may influence statistical measures such as the mean.

The outliers have been removed, resulting in a more compact and balanced distribution.

The dataset now better represents the majority of players and may be more suitable for certain machine learning models.

(Outliers in the *Overall Rating* variable were removed temporarily to examine their impact on the data distribution and visualizations. Since these observations represent real high-performing players rather than errors, the removal was conducted for practice and exploratory analysis purposes only.)



The correlation matrix shows that **Overall Rating** is most strongly associated with a player's technical ability, potential, and reactions.

The strongest predictor of a high overall rating is **Reactions (0.86)**. Players who react quickly to game situations tend to have significantly higher ratings.

Short Passing (**0.51**)

Vision (**0.51**)

Long Passing (**0.49**)

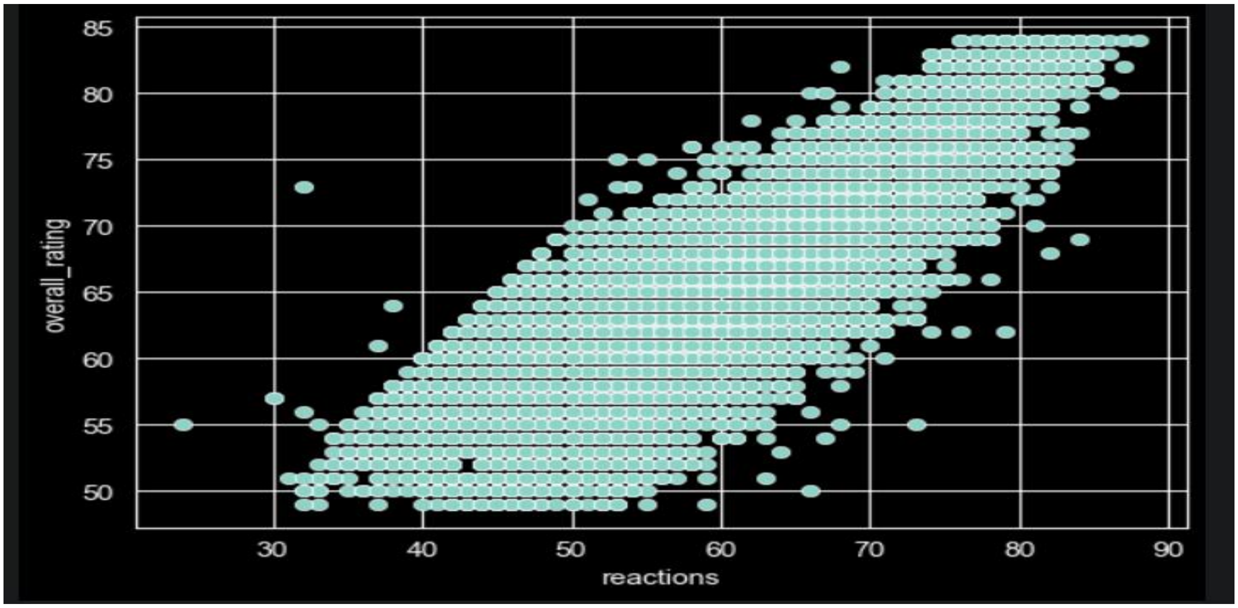
Ball Control (**0.46**)

Curve (**0.42**)

Crossing (**0.40**)

This indicates that players with strong technical and playmaking abilities generally receive higher overall ratings.

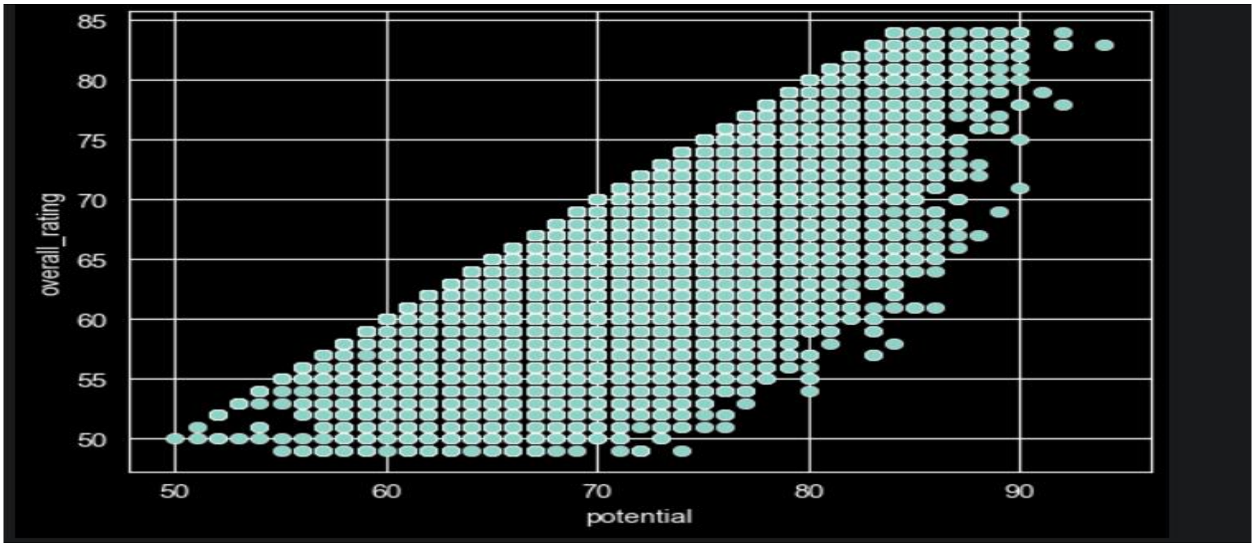
Pure speed and athleticism contribute less to overall rating than technical skills and game intelligence.



The scatter plot shows a **strong positive linear relationship** between **Reactions** and **Overall Rating**.

As **Reactions** increase, **Overall Rating** also tends to increase.

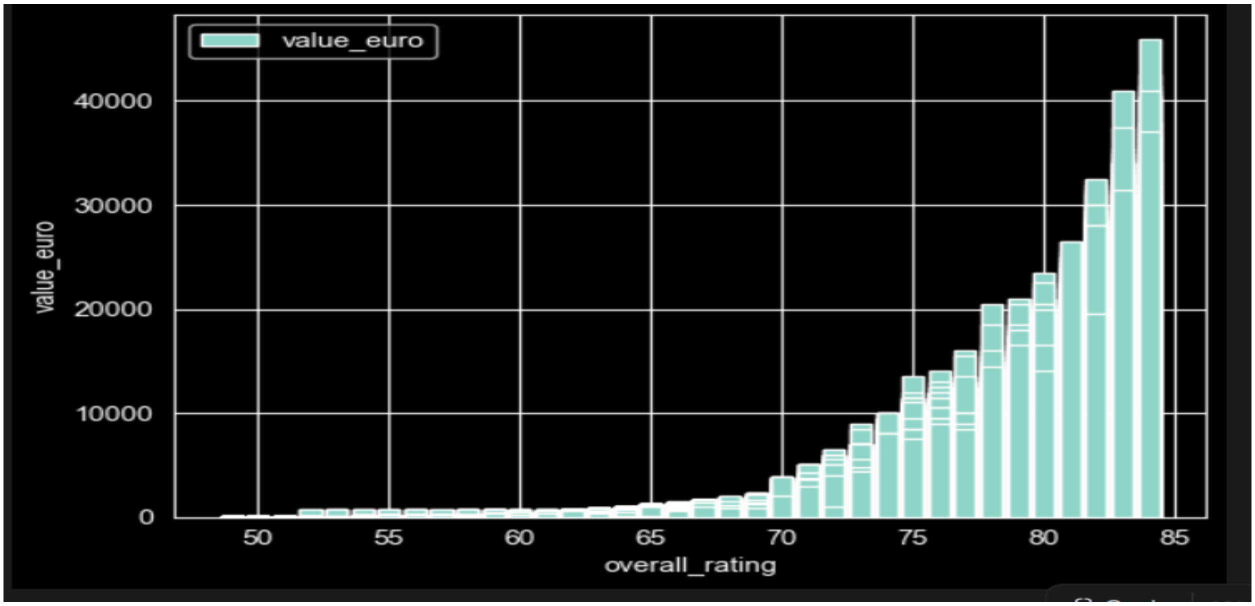
Players with low reaction scores generally have lower overall ratings.



The scatter plot indicates a **strong positive relationship** between **Potential** and **Overall Rating**.

As **Potential** increases, **Overall Rating** generally increases as well.

(Outliers were detected in the Potential variable using the IQR method. These observations represent highly talented players rather than data errors)

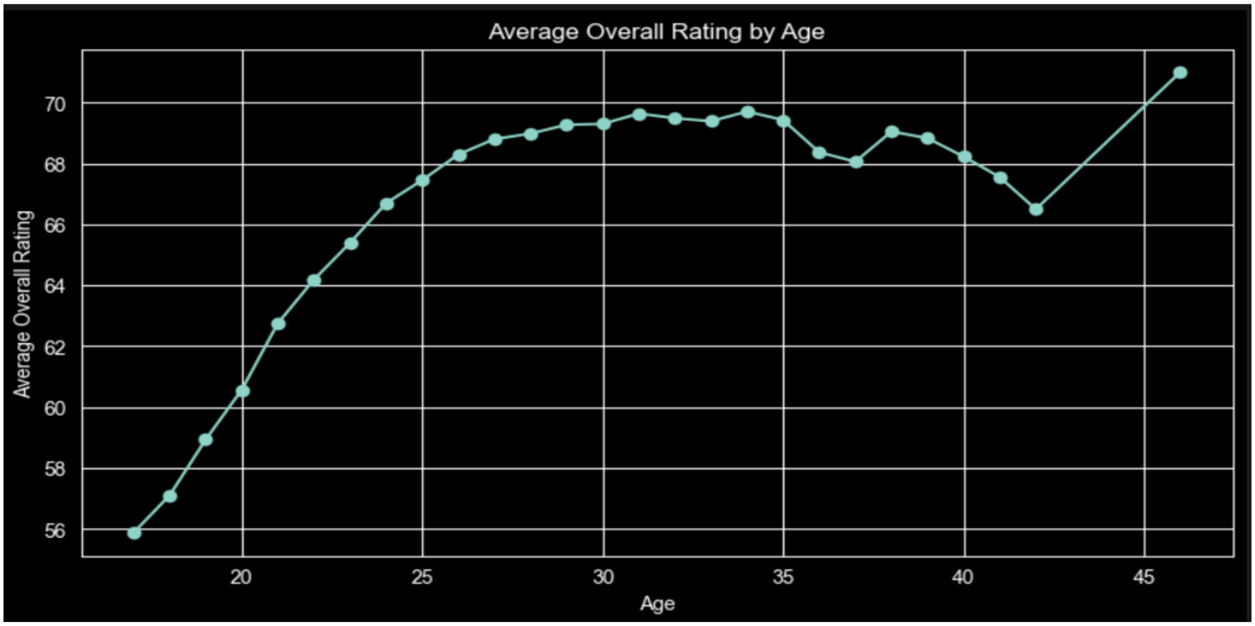


This plot shows a **strong positive relationship** between **Overall Rating** and **Player Market Value (€)**.

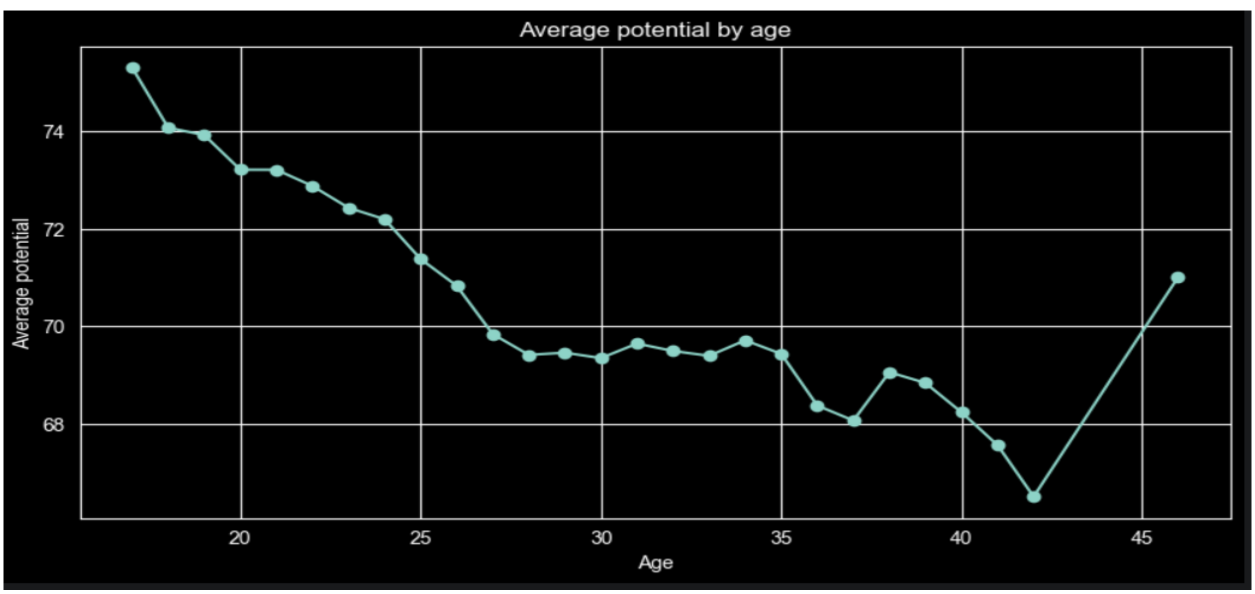
Player value increases as Overall Rating increases.

Between ratings **50–70**, market value grows slowly. After approximately **75 Overall Rating**, player value increases much more rapidly.

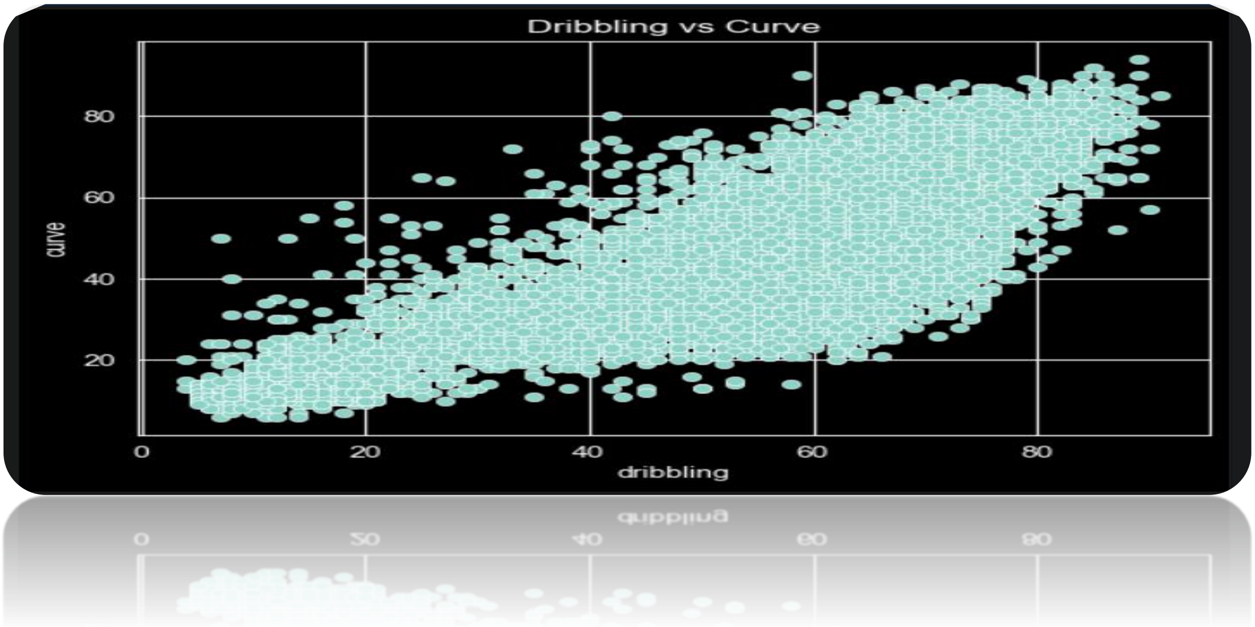
This pattern suggests that small improvements in rating at the elite level lead to disproportionately large increases in market value.



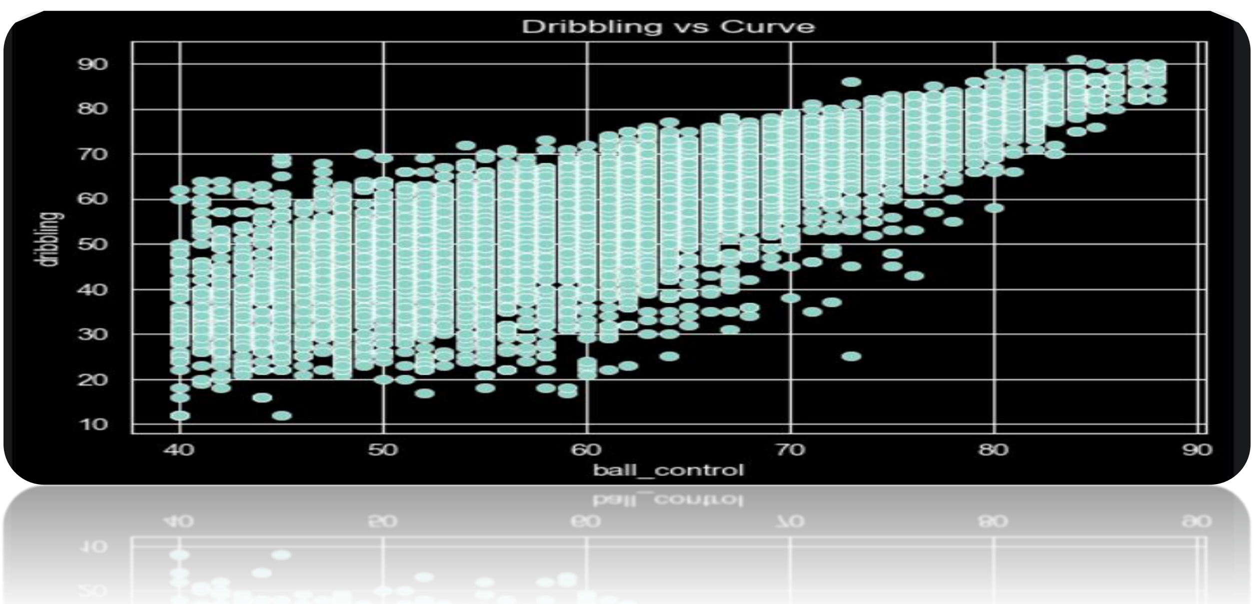
The results show that the average Overall Rating increases steadily from ages 17 to 30, indicating continuous player development and experience gain. Players reach their peak performance between ages 30 and 35, where the average rating is highest. After age 35, the average rating gradually declines. Large fluctuations at very high ages are likely due to the small number of players in those age groups.



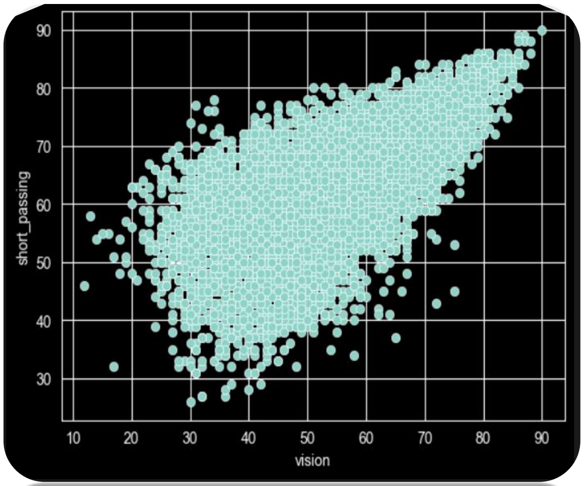
The results indicate that younger players tend to have higher potential, while average potential decreases as age increases. This suggests that younger players are expected to improve more in the future, whereas older players have less room for further development.

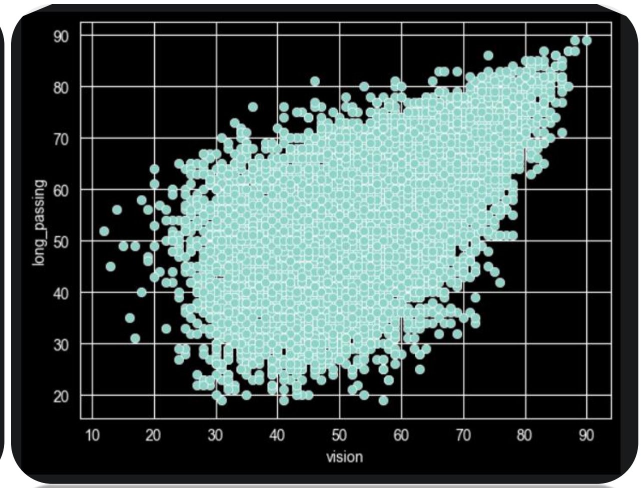


The scatter plot reveals a strong positive relationship between Dribbling and Curve. Players with better dribbling skills tend to have higher curve ratings, suggesting that these technical abilities are closely related.

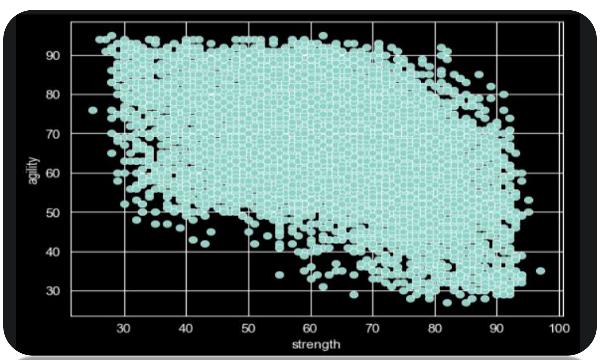


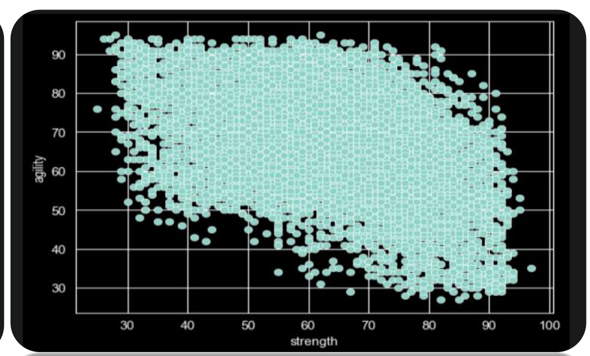
The scatter plot indicates a very strong positive relationship between Ball Control and Dribbling. Players with superior ball control generally demonstrate better dribbling ability, highlighting the close connection between these two technical skills





The analysis indicates that Vision is strongly associated with both Short Passing and Long Passing. Players with higher vision ratings generally demonstrate better passing performance, highlighting the importance of field awareness and decision-making in playmaking. The relationship is slightly stronger for Short Passing, suggesting that vision contributes more directly to accurate short-distance ball distribution.





The analysis reveals that stronger players generally exhibit lower Balance and Agility scores. This indicates a trade-off between physical power and mobility-related attributes. The effect is particularly noticeable for Balance, which shows the strongest negative correlation with Strength in the dataset.